Murra Rajesh Kumar Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.91
Nano Jr college	Class XII (SSC)	2023	97.60%
Montessori English Medium Primary School	Class X (CBSE)	2021	92.83%

WORK EXPERIENCE

Python Developmental Activity for Netlists and SAIF | python:

- Under guidance of Prof. Amit Acharya did some developmental activities in python for understanding and parsing netlist and SAIF files

SKILLS

- **Programming Languages :** C | Verilog | Python.
- **Tools Used :** MATLAB | LTspice | Modelsim | Quartus Prime Lite | OpenLane | GTKwave.
- **Spoken Languages :** Japanese, English, Telugu, Hindi.

RELEVANT COURSES

Core Courses : Circuits and Network Analysis, Digital Systems, Analog Electronics, Digital Signal Processing, Semiconductor Device Fundamentals, Electric Machines, Control Systems, Power Systems, Intro to CMOS, Engineering Electromagnetics.

Lab Courses: Electrical Circuits Lab, FPGA lab, Digital System Lab, Electronic Devices and Circuits Lab.

Computer Science Courses: Introduction to Programming, Scientific Programming for Electrical Engineers, Data Structures & Applications, Machine Learning.

Math Courses: Mathematics for Electrical Engineers, Matrix Theory, Differential Equations and Transform Techniques, Probability, Random Process.

PROJECTS

Solar PV Plant Performance & Asset Management Analysis | Python, Excel

- Acquired and processed high-resolution PV plant operational data from the NREL/OEDI PVDAQ database, including POA irradiance, DC/AC power, voltage, current, and temperature measurements.
- Developed a Python-based data-analysis pipeline to perform data-quality checks and evaluate PV system performance at yearly, monthly, and daily time scales.
- Analyzed energy generation, capacity factor, DC-AC conversion efficiency, irradiance-power characteristics, and operating anomalies to assess PV asset performance and identify potential losses.

Solar PV Operational Data Analysis | Excel, Python

- Processed one-minute resolution solar PV operational data and performed data-quality checks, time-series analysis, and KPI calculations using Excel.

- Analyzed irradiance, DC/AC power, voltage, current, and temperature measurements to evaluate daily plant operating performance.

- Built performance and anomaly-analysis worksheets to identify periods of abnormal generation relative to available solar irradiance and operating conditions.

- Developed Excel-based charts and KPI summaries for structured reporting of daily PV plant performance.

FFT & FIR Filter Accelerator on FPGA | Verilog, Python, Quartus Prime Lite, GTKWave, Icarus, DE1-SoC Cyclone V:

- Developed synthesizable Verilog implementations of an **8-point FFT** and a **101-tap FIR Low-Pass Filter** for digital signal processing applications.
- Verified RTL functionality using Verilog testbenches by comparing simulation outputs against Python-generated golden reference values.
- Implemented and synthesized the designs on the DE1-SoC cyclone V FPGA using Quartus Prime lite and Validated correct hardware functionality on FPGA board.
- Designed a ROM based input stimulus generator to supply identical test vectors used during Python-based verification, ensuring consistency between stimulus and FPGA results.

ECG Heart Rate (BPM) Calculator | Verilog, Python, ModelSim, GTKWave, OpenLane

- Developed a Verilog-Based ECG heart Rate (BPM) Calculator consisting of preprocessing filters, peak Detection, TIme interval measurement, and BPM computation modules.
- Designed and integrated a synchronous circular FIFO and Control Unit (CU) to buffer ECG samples from analog fron end and coordinate data transfer to the digital processing block.
- Verified RTL functionality using Verilog testbenches by comparing simulation outputs against Python-generated golden reference values and analyzed waveforms using ModelSima and GTKWave.
- Performed ASIC RTL synthesis using OpenLane flow, gaining practical exposure to synthesis, timing analysis, and standard-cell-based digital implementation.
- Completed as the major project for a VLSI Internship Training program and received certification upon successfull completion